Statistics — Day 2

Data Types · Central Tendency · Dispersion

01 April 2026 · Statistics Day-2

Data Types	Mean	Median	Mode	Variance & SD

§ 01 Data Types in Statistics

PROGRAMMING DATA TYPES VS STATISTICAL DATA TYPES

Programming types: int, float, string, boolean, list, set, datetime, dataframe (Python/Pandas)

Statistical types: How the data *behaves* — what operations make sense on it.

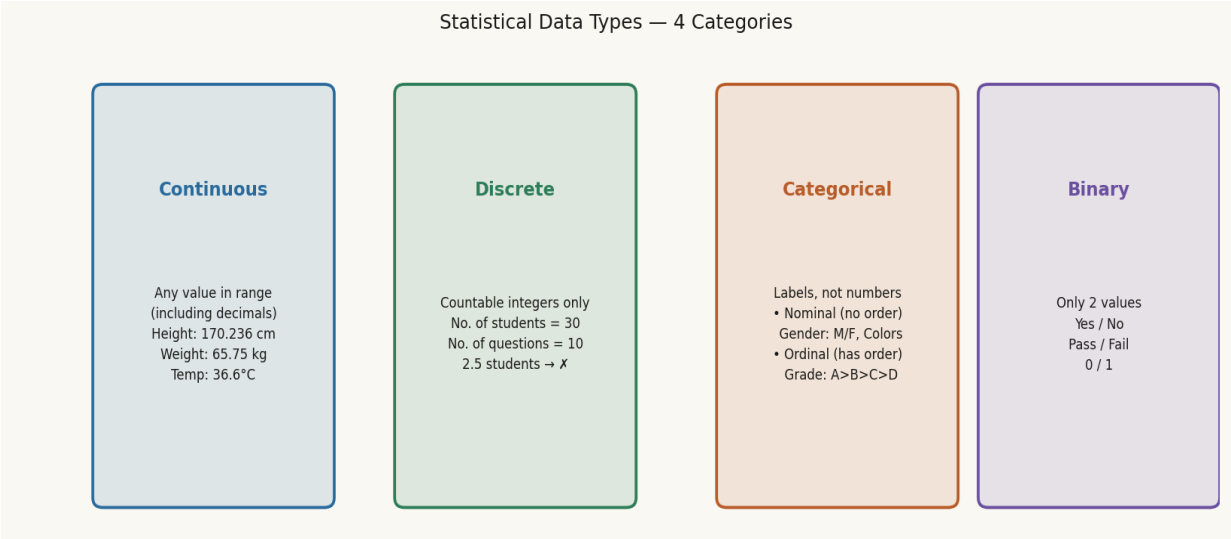

Figure 1: Four statistical data types — Continuous, Discrete, Categorical, Binary

Type	Definition	Key rule	Examples
Continuous	Any value in range, including decimals	Infinite precision possible	Height=170.236 cm, Weight=65.75 kg, Temp=36.6°C

Discrete	Countable integers only	No fractions — 2.5 students = ✗	No. of students=30, No. of questions=10
Categorical – Nominal	Labels with NO order	Can't rank them	Gender (M/F), Blood group (A/B/O), Colors (Red/Blue)
Categorical – Ordinal	Labels WITH order	Can rank but gaps aren't equal	Grade A>B>C>D, Satisfaction: Poor→Excellent, Pain: Mild→Severe
Binary	Only 2 values	Special case of categorical	Yes/No, Pass/Fail, 0/1, True/False

Student dataset example:

Student	Marks	Gender	Grade	Passed	Height	Weight
A	85	M	A	Yes	—	—
B	72	F	B	Yes	—	—
C	40	M	D	No	—	—

TYPE OF EACH COLUMN

Marks → **Continuous** (0–100, can be 72.5) | Gender → **Categorical Nominal** (no order)

Grade → **Categorical Ordinal** (A>B>C>D) | Passed → **Binary** (Yes/No)

Height/Weight → **Continuous** (decimals possible)

§ 02 Measurement of Central Tendency

GOAL

Find the **center** of the data — a single number that summarises the whole dataset.

Used for: aggregate data → summary → compare groups (A scored 60 → okay, B scored 85 → best, C scored 40 → poor)

Three measures: **Mean** (numerical) · **Median** (numerical) · **Mode** (categorical)

(1) Mean — Average

```
Mean = Σx■ / n = (x■ + x■ + ... + x■) / n
```

`Marks = [10, 20, 30] → Mean = (10+20+30)/3 = 60/3 =` **`20`**

DRAWBACK OF MEAN — SENSITIVE TO OUTLIERS

A → [10,20,40,50] → Mean=30 | B → [10,20,40,90] → Mean=40

C → [10,20,90,60] → Mean=32.5 | D → [10,20,40,**500**] → Mean=142.5

→ Mean is **shifted by the large number** (500). Because of 500, mean=142.5 is misleading!

→ Mean does NOT represent the true center when outliers are present.

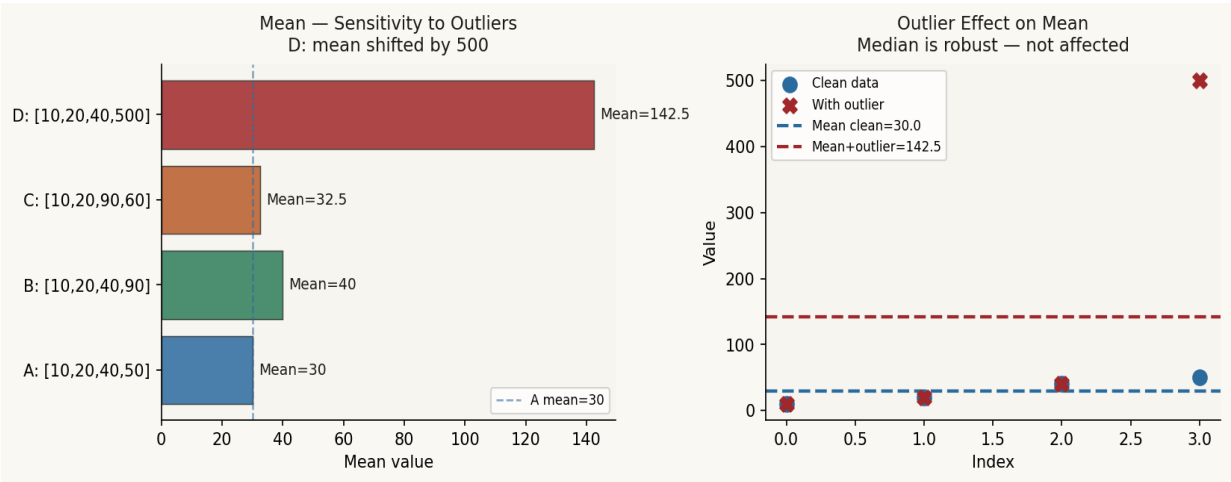

Figure 2: Mean sensitivity to outliers — dataset D with 500 shifts mean to 142.5

(2) Median — Middle Value

DEFINITION & RULE

Sort the data first. Then find the middle value.

→ **Odd n:** take the middle element directly

→ **Even n:** take average of the 2 middle values

Median is **robust** — not affected by outliers!

```
Step 1: Sort data
Step 2: If n is odd → Median = value at position (n+1)/2
If n is even → Median = average of values at positions n/2 and n/2+1
```

Example (n=9, odd): A = [60,50,40,30,10,50,40,60,55]

`sort(A) = [10, 30, 40, 40, 50, 50, 55, 60, 60] → Median = 50 (5th value)`

`Example with outlier:` `sort = [10,30,40,40,50,50,55,60,8000] → Median still = 50`

Mean would be (10+30+40+40+50+50+55+60+8000)/9 ≈ 926 — completely misleading!

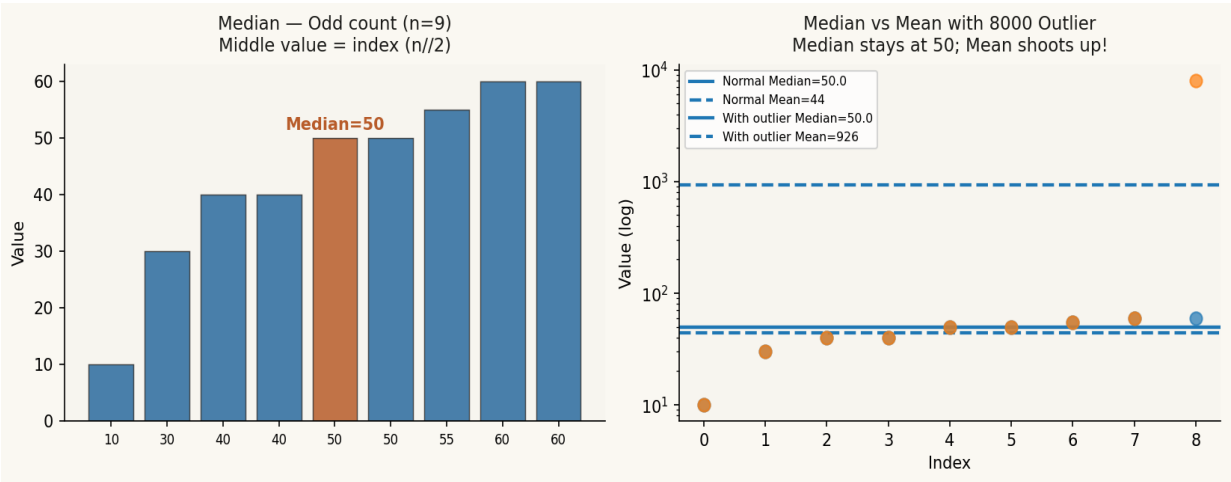

Figure 3: Median — middle value with odd count, and robustness vs mean with outlier 8000

WHEN TO USE MEAN VS MEDIAN

→ Want aggregation showing data behaviour → **Mean or Median**

→ Want true center of data → **Mean or Median**

→ Data has outliers → **Median** (robust, not shifted)

→ Data is clean (no outliers) → **Mean** (uses all values)

Q: [10, 20, 30, 1000] → center = **Median** (mean is sensitive to 1000)

(3) Mode — Most Frequent Value

DEFINITION

Mode = the value that appears most frequently.

Used for **categorical data** — mean/median don't make sense for categories.

Also works for numerical data → most frequent number.

Example — Sports: who won the most games?

`Results: A, B, A, B, A, B, C, C, C, C → A:3, B:3, C:4 → Mode = C`

Example — Marks 1–10 scale (60 students):

`A:1, B:2, C:7, D:7, F:8 → Mode = F (most students scored F)`

MODE USE CASES

• Numerical → most frequent *number*

• Categorical → most frequent *category*

Mean and Median don't work for categorical (can't average 'Male' and 'Female')

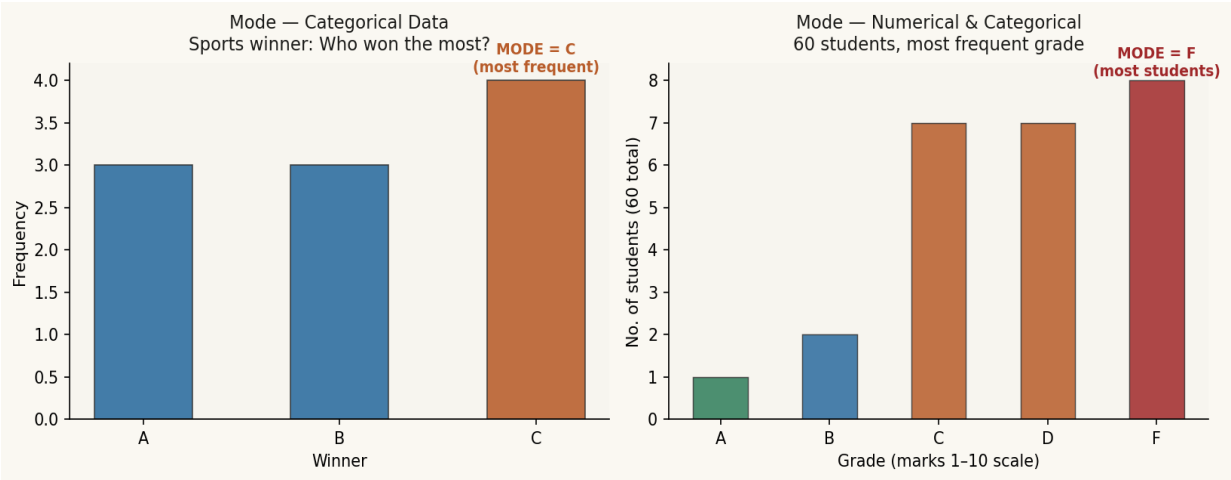

Figure 4: Mode — categorical sports winner (C wins most) and grade distribution

§ 03 Measurement of Dispersion (Spread)

GOAL

Find **how spread out** the data is — the common spread / average deviation from center.

Spread = Standard Deviation (SD). Two students can have the same mean but different spread.

A → [50,51,49,50,52] → Mean=50.4 (consistent) | B → [0,100,10,90,50] → Mean=50 (unpredictable)

Both have ~same mean but A is **stable**, B is **unstable** (high spread).

Variance — The Building Block

```
Variance = Σ (x■ − μ)² / n (σ² — lower case sigma squared)
= average of (difference of each point from mean)²
Why square? → removes directional sign (+/-) · keeps magnitude
```

Worked Example — [10, 20, 30]:

Step	Calculation	Value
Mean (μ)	(10+20+30)/3	= 20

Deviation of 10	$(10–20)^2 = (–10)^2$	= 100
Deviation of 20	$(20–20)^2 = (0)^2$	= 0
Deviation of 30	$(30–20)^2 = (10)^2$	= 100
Variance (σ^2)	(100+0+100)/3 = 200/3	≈ 66.67 units²
SD (σ)	$\sqrt{66.67}$	≈ 8.16 units

Standard Deviation = √Variance = $\sqrt{(\sigma^2)} = \sigma$
SD is in the SAME UNIT as the data (variance = unit², SD = unit)
Series [10,20,30]: numbers deviate from mean (20) by 8.16 on average

Interpretation: [10, 20, 30] → numbers are 11.84, 20, 28.16 (mean±SD range)

VARIANCE VS SD — WHY WE PREFER SD

Variance → units² (e.g. inch²) — hard to interpret compared to original data

SD → same unit as data (e.g. inch) — directly comparable

SD = √Variance → converts back to original unit → easier to understand

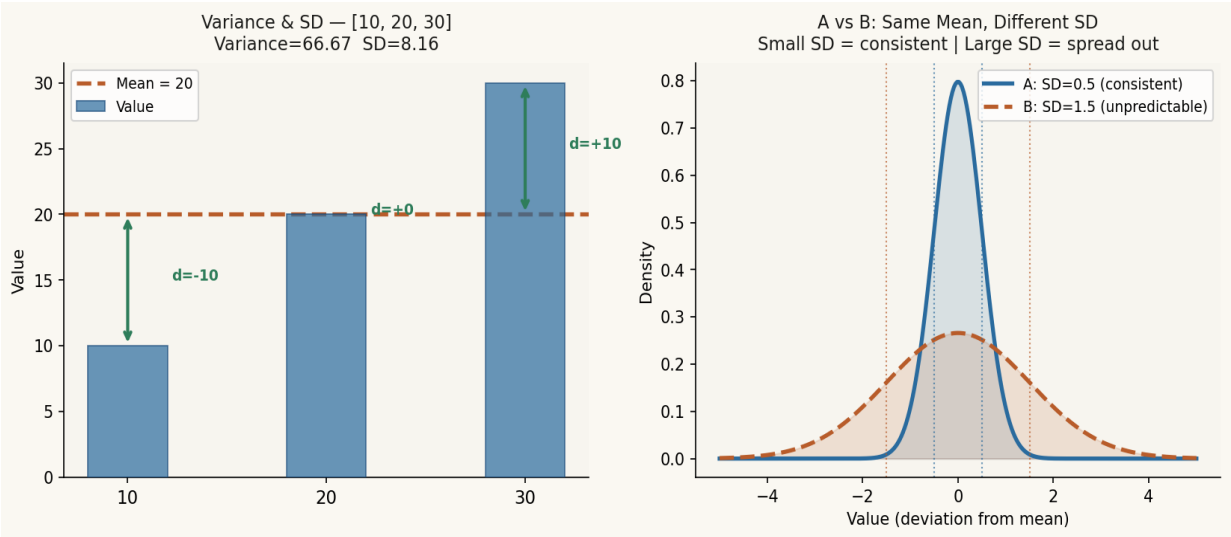

Figure 5: Variance/SD step-by-step for [10,20,30] and A vs B with same mean, different SD

§ 04 Summary — All Formulas

Measure	Formula	Best for	Drawback
Mean	$\Sigma x_i / n$	Clean numerical data	Sensitive to outliers
Median	Middle value (sorted)	Data with outliers	Ignores extreme values
Mode	Most frequent value	Categorical data	Multiple modes possible
Variance	$\Sigma(x_i-\mu)^2 / n$	Measuring spread	Units are squared (unit²)
Std Dev	$\sqrt{\text{Variance}} = \sqrt{(\Sigma(x_i-\mu)^2/n)}$	Spread in original units	Affected by outliers

Statistics Notes · 01 Apr 2026 · Data Types · Mean · Median · Mode · Variance · Standard Deviation